

Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

Next up:

- Midterms returned by class time Friday.
- This week: Chapter 4
- Quiz 8 on Friday. (Computing determinants.)
- Problem session Friday, 3-5pm in Ryerson 177.
- ***Update:*** Final three homeworks (written and WeBWorK) due on Mondays at 11:59pm. (Three days later than previously.)
- Sunday HW₅ Debrief (“Prebrief”) Cobb 115, 3-4pm

Second part of the course:

Just two new quizzes after Friday's Quiz 8 (on determinants)

Just 1.5 more homeworks after HW₅ (due Monday)

Second part of the course:

Just two new quizzes after Friday's Quiz 8 (on determinants)

Just 1.5 more homeworks after HW₅ (due Monday)

Less new vocabulary, less new skills.

More big results, more theory, emphasis on matrix decompositions.

Second part of the course:

Just two new quizzes after Friday's Quiz 8 (on determinants)

Just 1.5 more homeworks after HW5 (due Monday)

Less new vocabulary, less new skills.

More big results, more theory, emphasis on matrix decompositions.

Read the book but don't get overwhelmed by the details.

Solidify your understanding of the first part of the course.

Revisit your flex plan and goals.

Ch 4

eigenvalues: “local stretch factors” of a linear transformation

eigenvectors: vectors that a linear transformation acts upon by scaling

Ch 4

eigenvalues: “local stretch factors” of a linear transformation

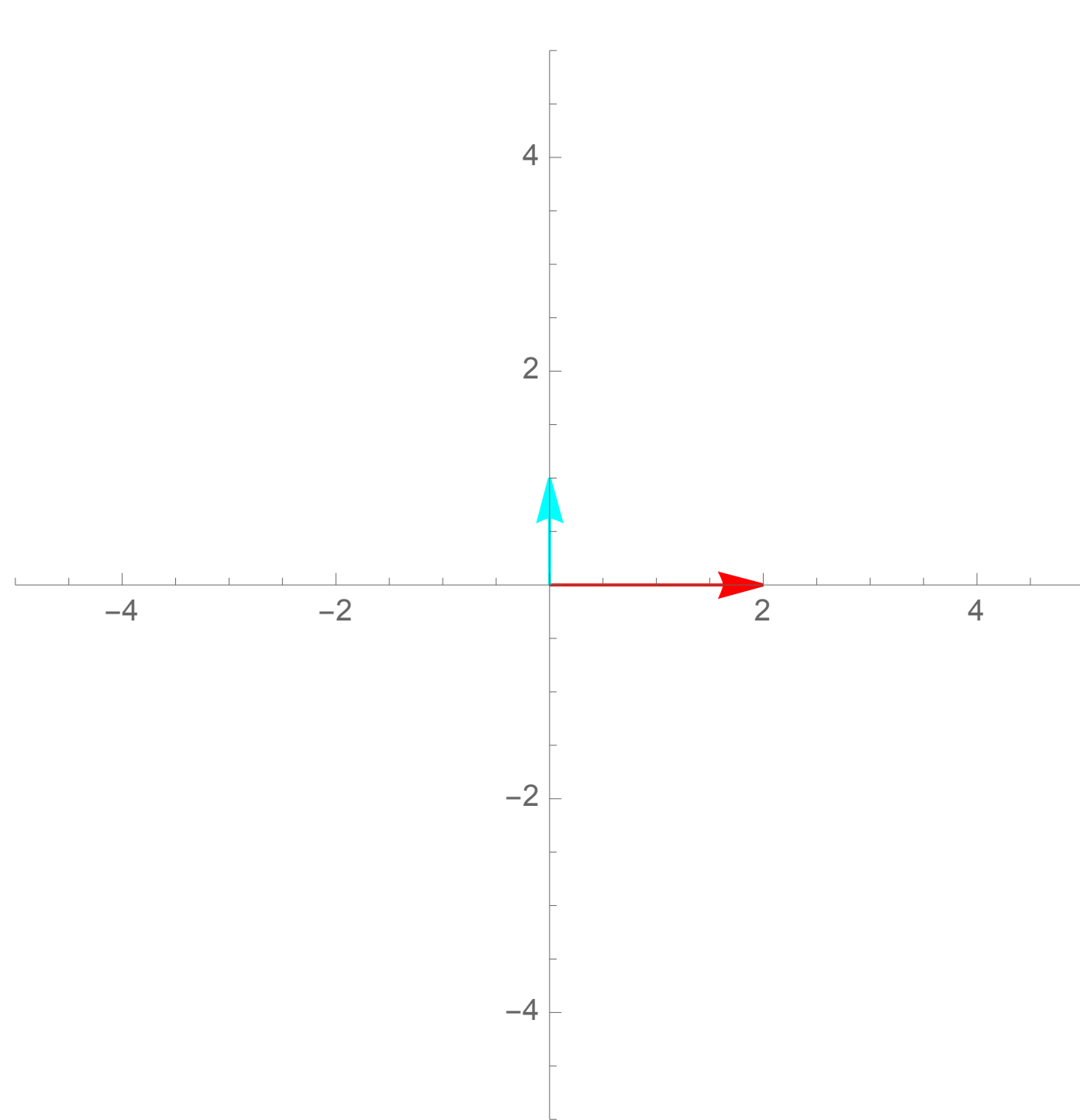
eigenvectors: vectors that a linear transformation acts upon by scaling

“eigen” is German and means something like
“belonging to” or “characteristic”

Ch 4

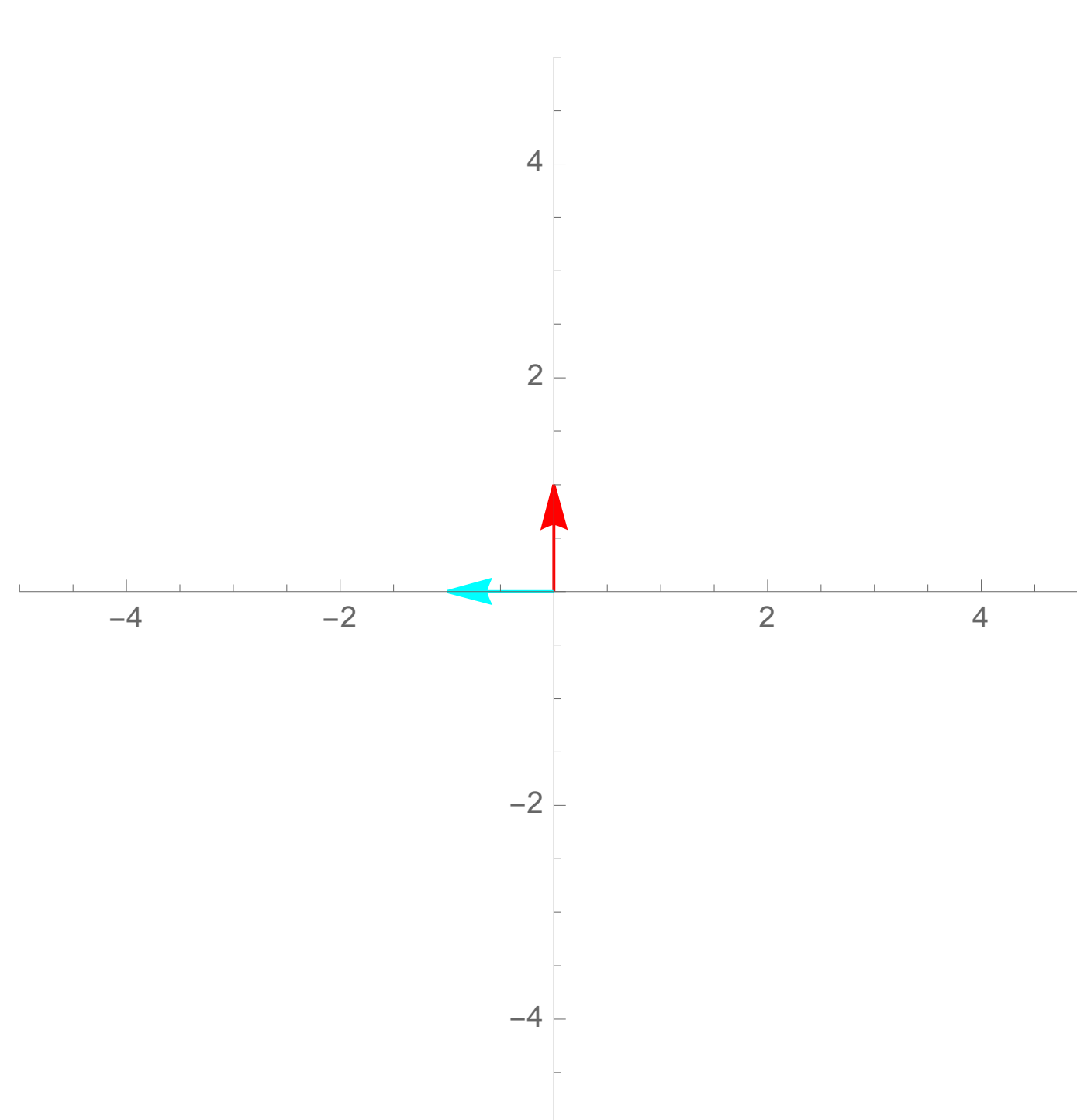
eigenvalues: “local stretch factors” of a linear transformation

eigenvectors: vectors that a linear transformation acts upon by scaling



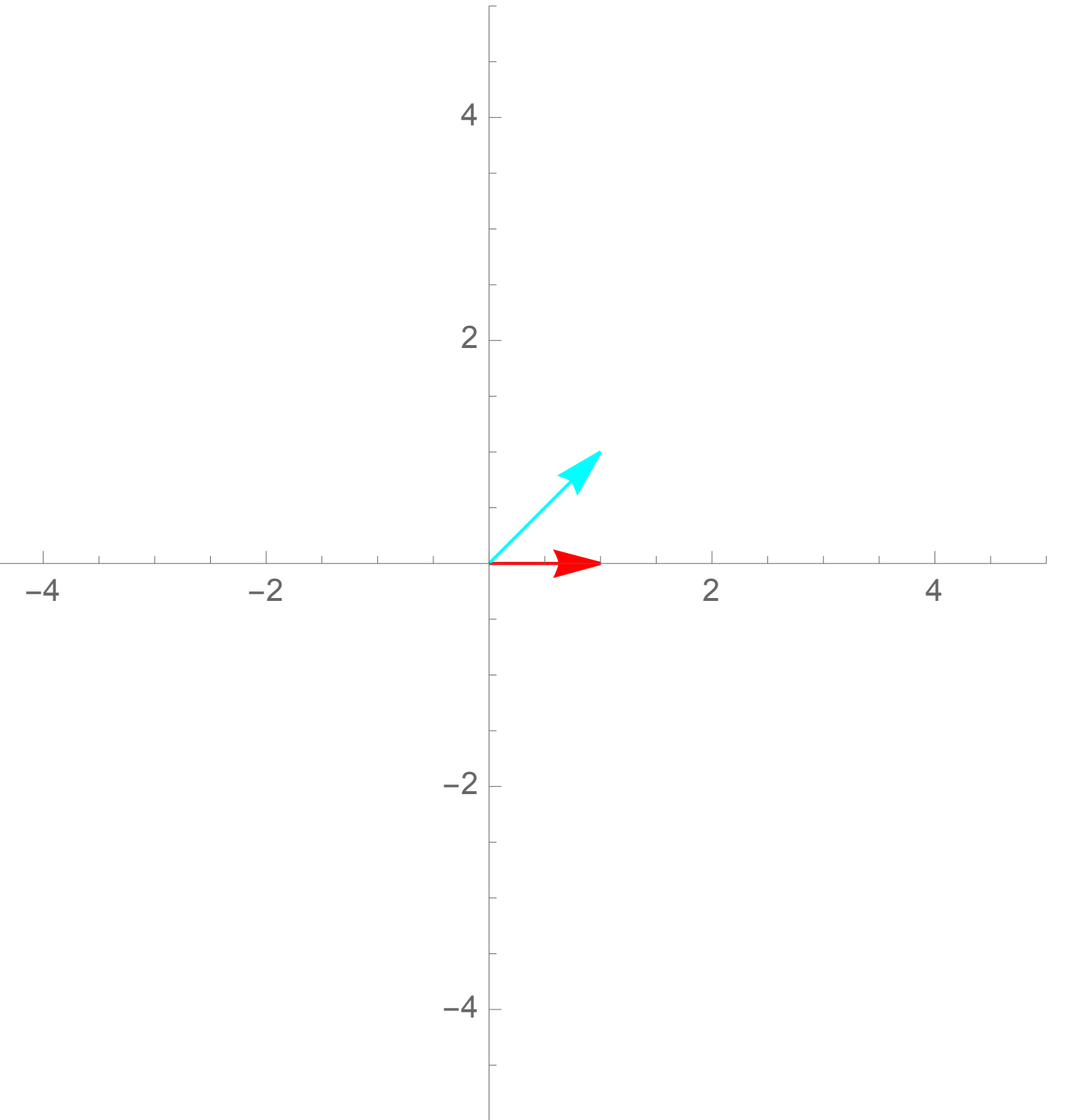
$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal



$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

rotation

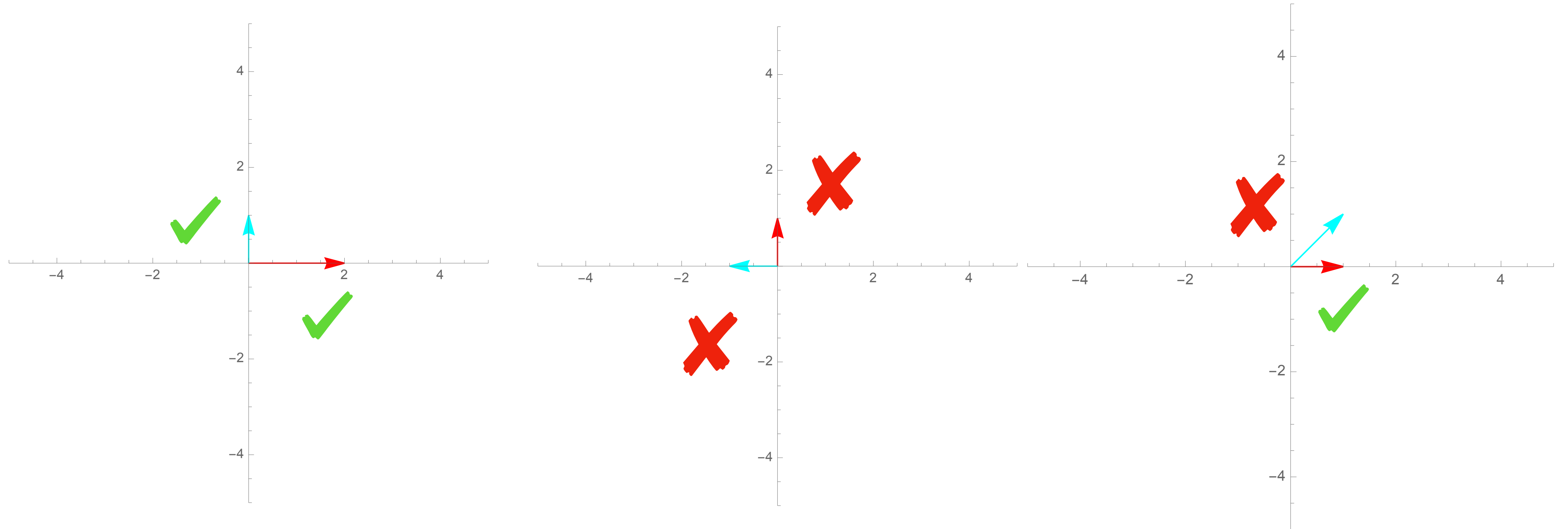


$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

shear

Ch 4

eigenvalues: “local stretch factors” of a linear transformation
eigenvectors: vectors that a linear transformation acts upon by scaling



$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

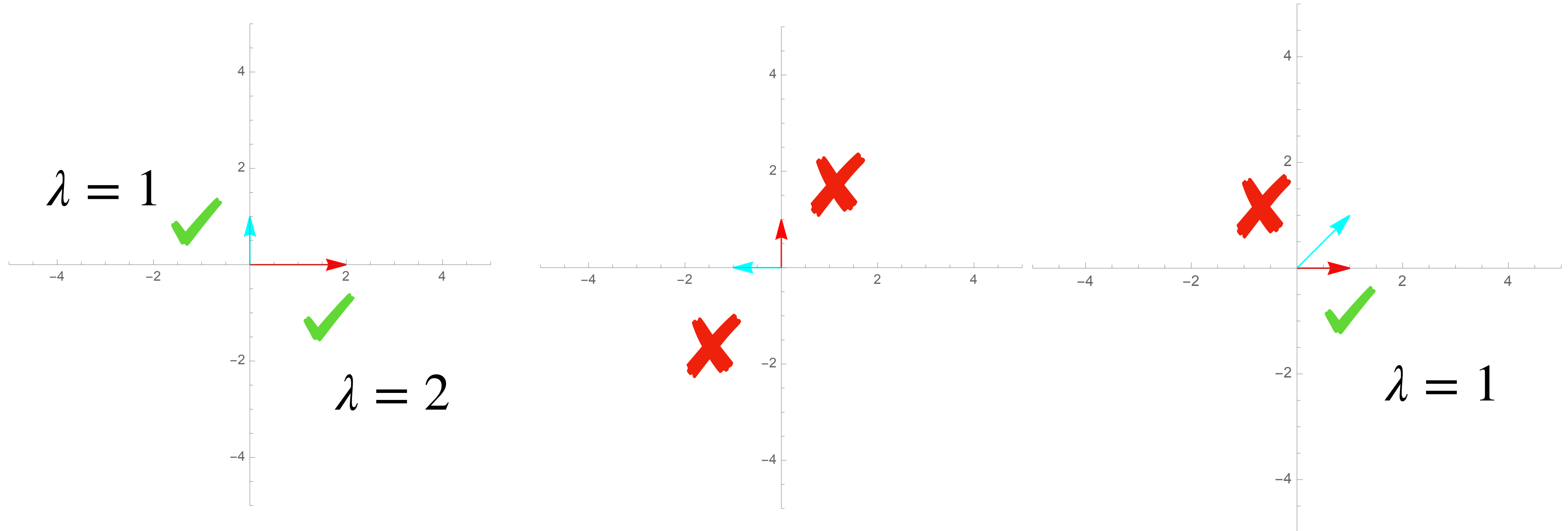
rotation

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

shear

Ch 4

eigenvalues: “local stretch factors” of a linear transformation
eigenvectors: vectors that a linear transformation acts upon by scaling



$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

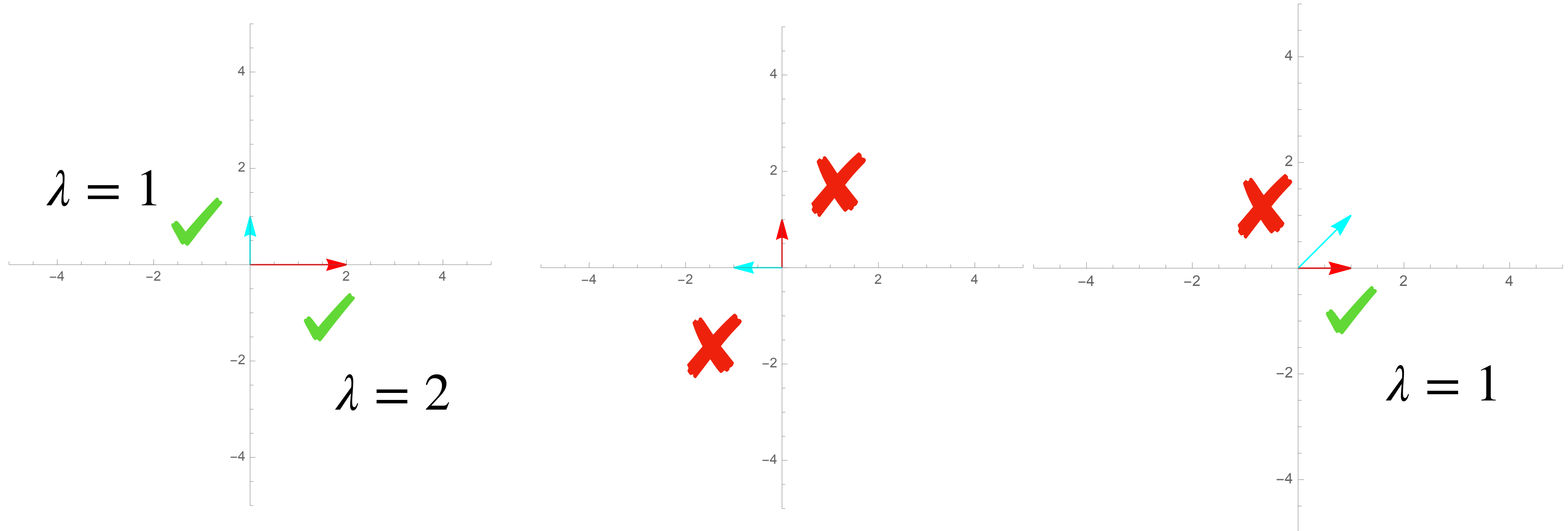
rotation

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

shear

Ch 4

eigenvalues: “local stretch factors” of a linear transformation
eigenvectors: vectors that a linear transformation acts upon by scaling



$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

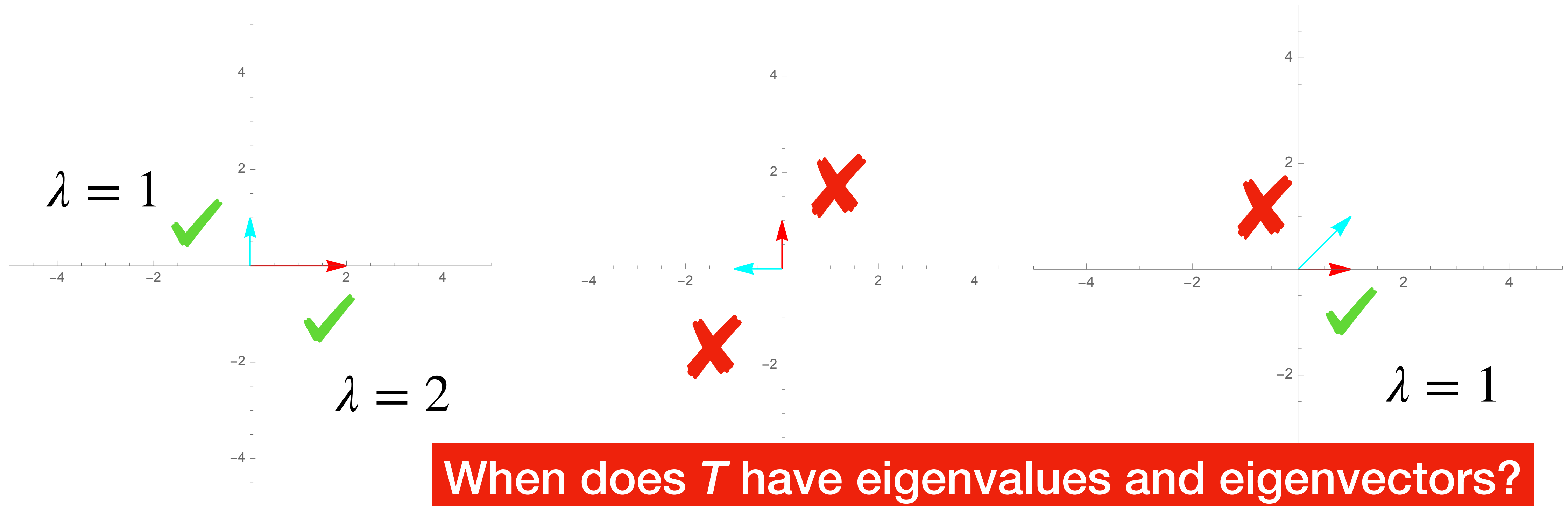
$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

shear

Note: eigenvectors don't have to be standard basis vectors.

Ch 4

eigenvalues: “local stretch factors” of a linear transformation
eigenvectors: vectors that a linear transformation acts upon by scaling



$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

When does T have eigenvalues and eigenvectors?

When it does, how can we find them?

What info do they give us about T ?

What properties do/can/must they have?

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

If v is an eigenvector, so is av for any nonzero scalar a .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

If v is an eigenvector, so is av for any nonzero scalar a .

If v and w are an eigenvectors with eigenvalue λ ,
so is every nonzero linear combination $av+bw$.

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

If v is an eigenvector, so is av for any nonzero scalar a .

If v and w are an eigenvectors with eigenvalue λ ,
so is every nonzero linear combination $av+bw$.

eigenspace: a subspace where every nonzero vector is an eigenvector
with some fixed eigenvalue λ .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

eigenspace: a subspace where every nonzero vector is an eigenvector
with some fixed eigenvalue λ .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

λ is an eigenvalue of $A \Leftrightarrow A - \lambda I$ has a nontrivial nullspace

6 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix
- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
 $A=LU$ and $A=PLU$ and $A=PLDU$ factorizations
- Calculating $\det A$

7 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix
- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
 $A=LU$ and $A=PLU$ and $A=PLDU$ factorizations
- Calculating $\det A$
- Finding eigenvalues and eigenvectors of A .

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

λ is an eigenvalue of $A \Leftrightarrow A - \lambda I$ has a nontrivial nullspace

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

$$\det(A - \lambda I) = 0$$



λ is an eigenvalue of $A \Leftrightarrow A - \lambda I$ has a nontrivial nullspace

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

$$Av = \lambda v$$

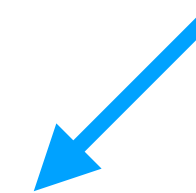
$$Av - \lambda v = 0$$

$$Av - (\lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

$$\det(A - \lambda I) = 0$$

characteristic
polynomial



λ is an eigenvalue of $A \Leftrightarrow A - \lambda I$ has a nontrivial nullspace

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

eigenvalue:
local stretch factor

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

eigenvalue:
local stretch factor

to find eigenvalues and eigenvectors and eigenspaces

$$(Av = \lambda v)$$

$$(AW = \lambda W)$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$. The scalar λ is the **eigenvalue** for the eigenvector v .

eigenvalue:
local stretch factor

to find eigenvalues and eigenvectors and eigenspaces

1. compute $\det(A - \lambda I)$

$$(Av = \lambda v)$$
$$(AW = \lambda W)$$

2. solve $\det(A - \lambda I) = 0$

3. find the nullspace of $A - \lambda_i I$ for each solution λ_i

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

eigenvalue:
local stretch factor

to find eigenvalues and eigenvectors and eigenspaces

-  1. compute $\det(A - \lambda I)$
2. solve $\det(A - \lambda I) = 0$
-  3. find the nullspace of $A - \lambda_i I$ for each solution λ_i

$$(Av = \lambda v)$$
$$(AW = \lambda W)$$

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.
The scalar λ is the **eigenvalue** for the eigenvector v .

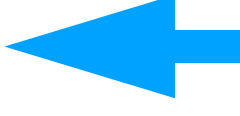
eigenvalue:
local stretch factor

to find eigenvalues and eigenvectors and eigenspaces



1. compute $\det(A - \lambda I)$

$$(Av = \lambda v)$$
$$(AW = \lambda W)$$

2. solve $\det(A - \lambda I) = 0$  The determinant is a polynomial in the entries of the matrix.



3. find the nullspace of $A - \lambda_i I$ for each solution λ_i

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1)$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

$$\lambda^2 - 10\lambda + 25 = 0$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

$$\lambda^2 - 10\lambda + 25 = 0 \quad \implies (\lambda - 5)(\lambda - 5) = 0 \quad \implies \lambda = 5$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

$$\lambda^2 - 10\lambda + 25 = 0 \quad \implies (\lambda - 5)(\lambda - 5) = 0 \quad \implies \lambda = 5$$

Find null $(A - 5I)$:

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

$$\lambda^2 - 10\lambda + 25 = 0 \quad \implies (\lambda - 5)(\lambda - 5) = 0 \quad \implies \lambda = 5$$

$$\text{Find null } (A - 5I) : \quad \begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix}$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

$$\lambda^2 - 10\lambda + 25 = 0 \quad \implies (\lambda - 5)(\lambda - 5) = 0 \quad \implies \lambda = 5$$

Find null $(A - 5I)$:

$$\begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix} \xrightarrow{\text{blue arrow}} \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix}$$

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$

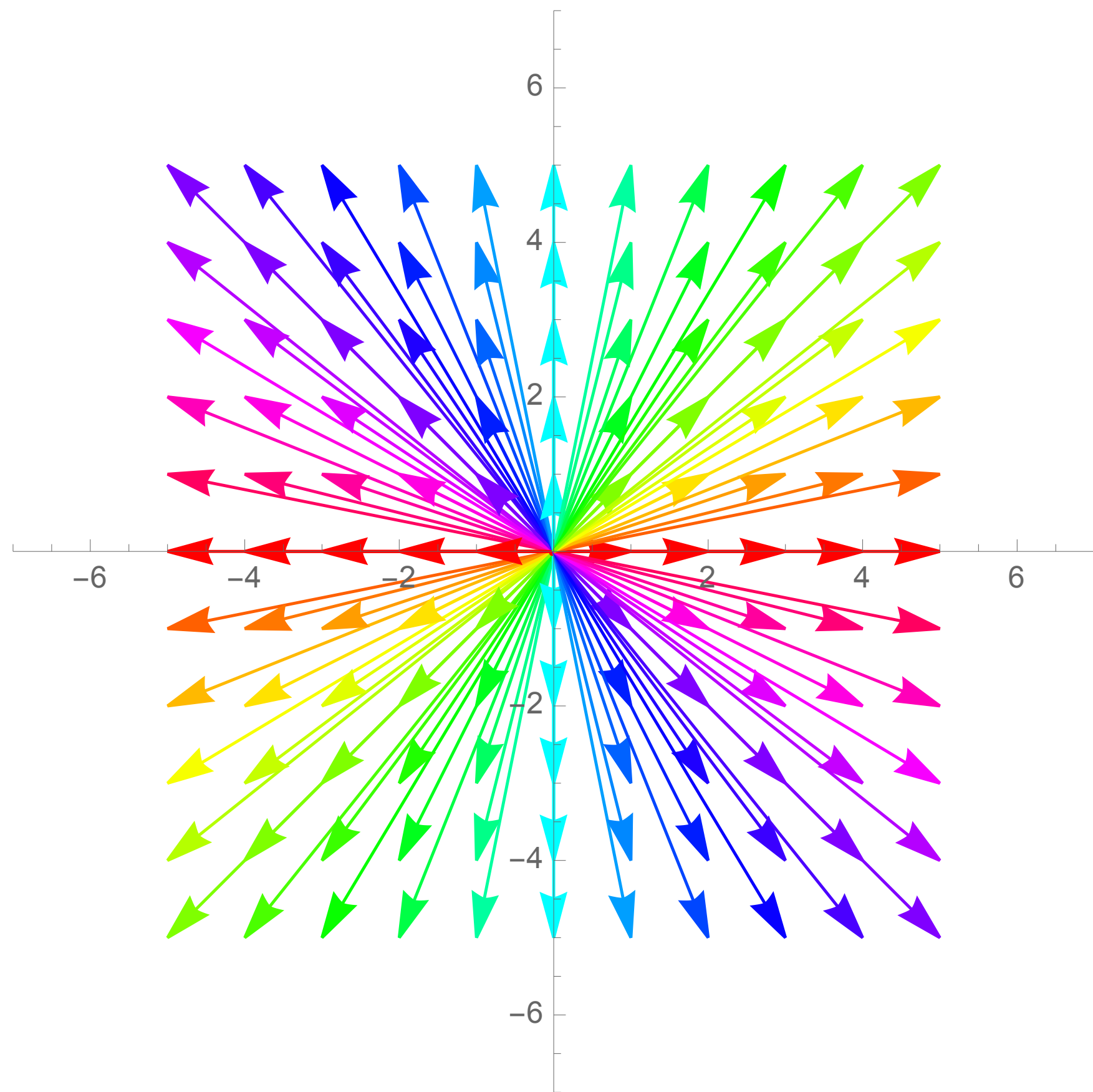
Find the eigenvalues and eigenvectors of $\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$

$$A = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} 7 - \lambda & -1 \\ 4 & 3 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) - (4 \cdot -1) = \lambda^2 - 10\lambda + 25$$

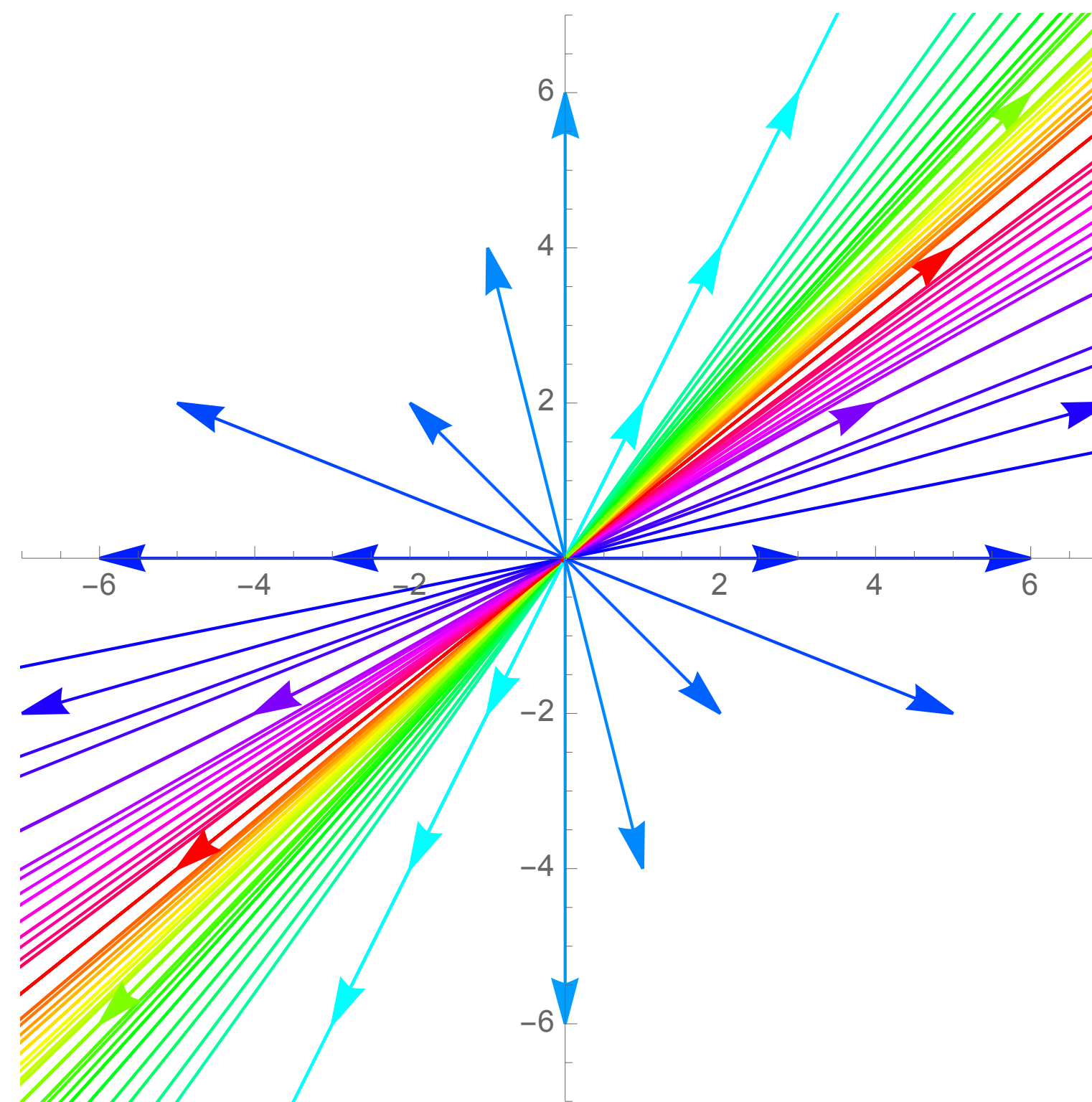
$$\lambda^2 - 10\lambda + 25 = 0 \quad \implies (\lambda - 5)(\lambda - 5) = 0 \quad \implies \lambda = 5$$

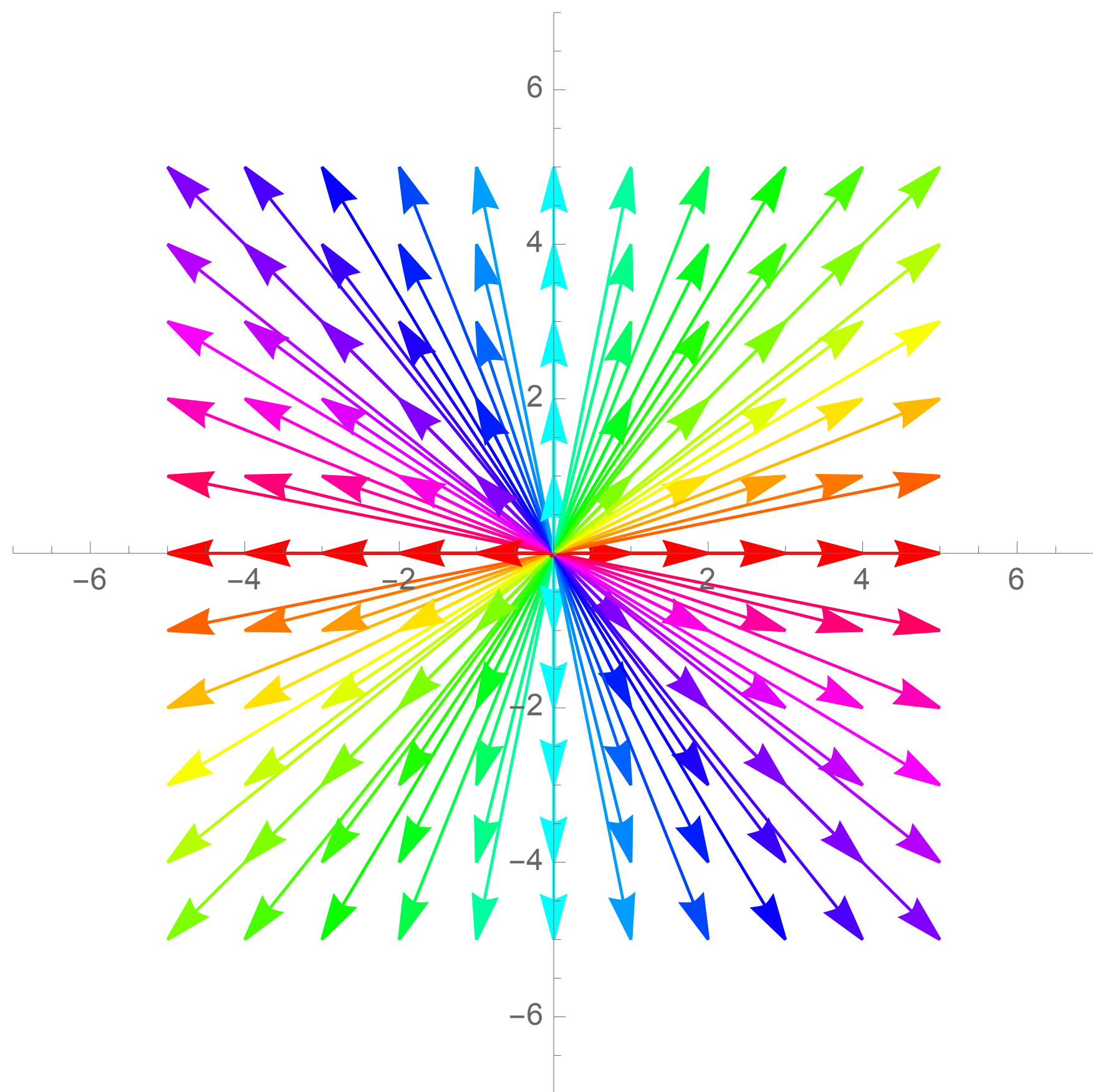
Find null $(A - 5I)$: $\begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix} \xrightarrow{\text{blue arrow}} \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix}$ So $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is a basis for the 5-eigenspace of A .



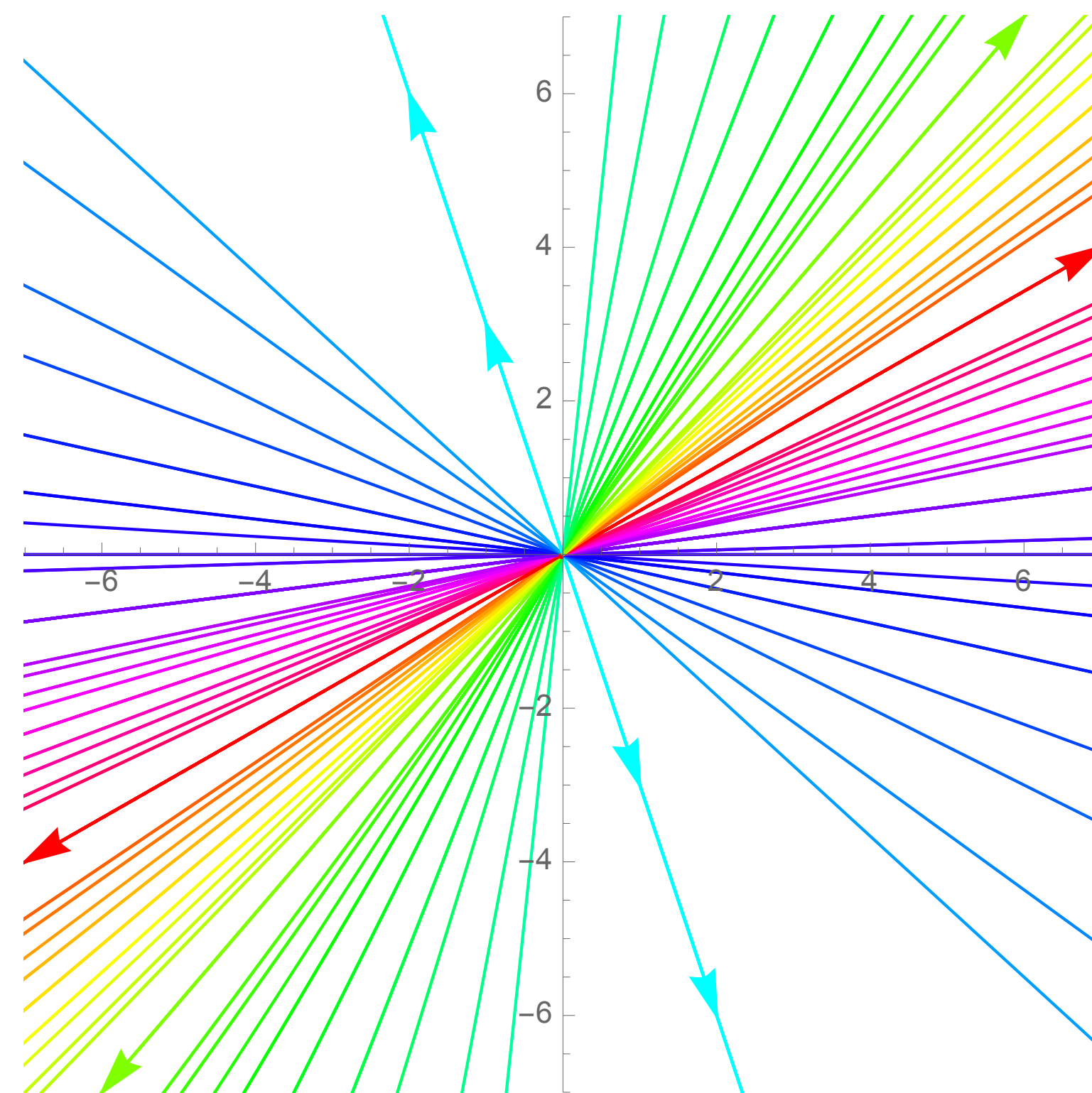
$$\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}$$

A large black arrow points from the first plot to the second plot, indicating a transformation.





$$\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$$



Next up:

- Midterms returned by class time Friday.
- This week: Chapter 4
- Quiz 8 on Friday. (Computing determinants.)
- Problem session Friday, 3-5pm in Ryerson 177.
- ***Update:*** Final three homeworks (written and WeBWorK) due on Mondays at 11:59pm. (Three days later than previously.)
- Sunday HW₅ Debrief (“Prebrief”) Cobb 115, 3-4pm